In [4]:
```
import pandas as pd

ratings = pd.read_csv("C:\\Users\\girna\\OneDrive\\Desktop\\recommendation-system\\
movies = pd.read_csv("C:\\Users\\girna\\OneDrive\\Desktop\\recommendation-system\\d

print(ratings.head())
print(movies.head())
```

```
   userId  movieId  rating  timestamp
0       1        1     4.0  964982703
1       1        3     4.0  964981247
2       1        6     4.0  964982224
3       1       47     5.0  964983815
4       1       50     5.0  964982931
   movieId                               title  \
0        1                    Toy Story (1995)
1        2                      Jumanji (1995)
2        3             Grumpier Old Men (1995)
3        4            Waiting to Exhale (1995)
4        5  Father of the Bride Part II (1995)

                                        genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                 Adventure|Children|Fantasy
2                            Comedy|Romance
3                       Comedy|Drama|Romance
4                                     Comedy
```

In [3]:
```
import os
print(os.getcwd())
```

```
C:\Users\girna
```

In [5]:
```
ratings.info()
movies.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100836 entries, 0 to 100835
Data columns (total 4 columns):
 #   Column     Non-Null Count   Dtype
---  ------     --------------   -----
 0   userId     100836 non-null  int64
 1   movieId    100836 non-null  int64
 2   rating     100836 non-null  float64
 3   timestamp  100836 non-null  int64
dtypes: float64(1), int64(3)
memory usage: 3.1 MB
<class 'pandas.DataFrame'>
RangeIndex: 9742 entries, 0 to 9741
Data columns (total 3 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   movieId  9742 non-null   int64
 1   title    9742 non-null   str
 2   genres   9742 non-null   str
dtypes: int64(1), str(2)
memory usage: 228.5 KB
```

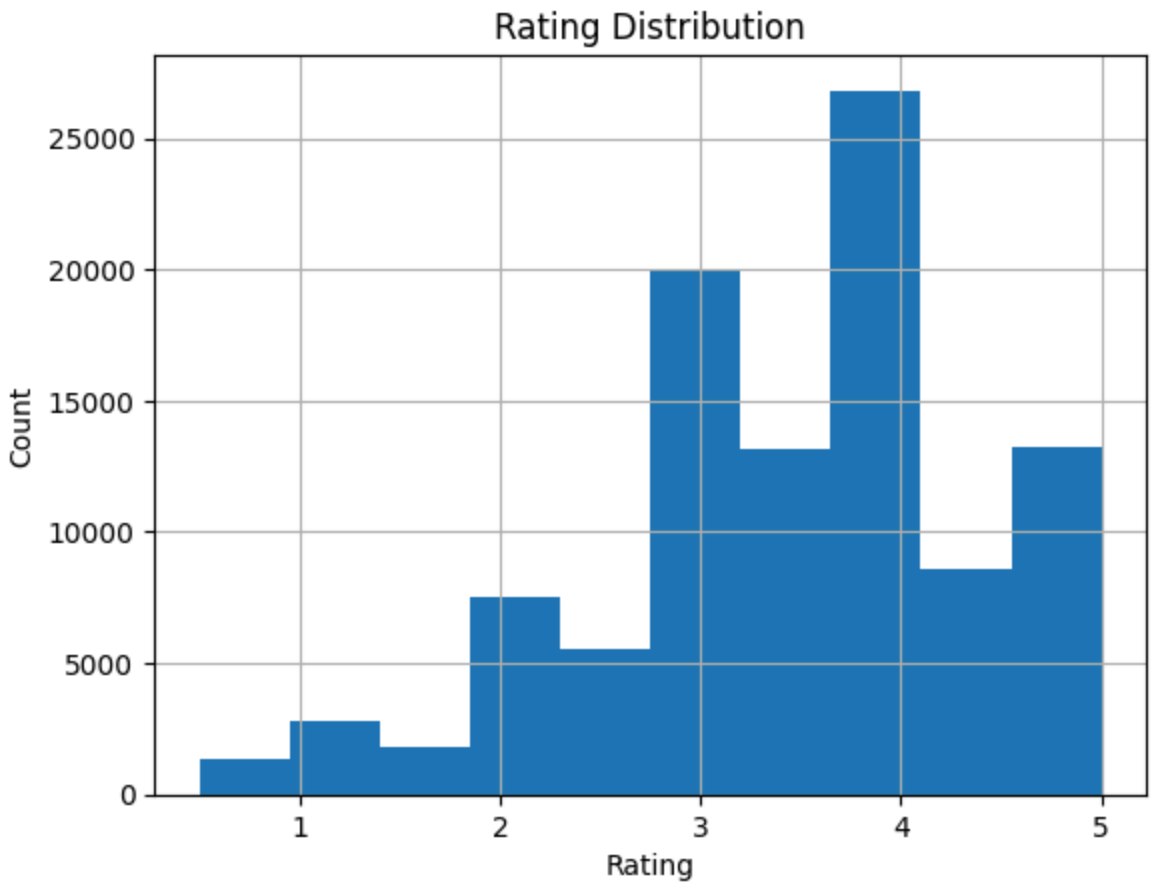
In [6]:
```
ratings.describe()
```

Out[6]:

	userId	movieId	rating	timestamp
count	100836.000000	100836.000000	100836.000000	1.008360e+05
mean	326.127564	19435.295718	3.501557	1.205946e+09
std	182.618491	35530.987199	1.042529	2.162610e+08
min	1.000000	1.000000	0.500000	8.281246e+08
25%	177.000000	1199.000000	3.000000	1.019124e+09
50%	325.000000	2991.000000	3.500000	1.186087e+09
75%	477.000000	8122.000000	4.000000	1.435994e+09
max	610.000000	193609.000000	5.000000	1.537799e+09

In [7]:
```
import matplotlib.pyplot as plt

ratings['rating'].hist()
plt.title("Rating Distribution")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```



In [8]:
```
top_movies = ratings.groupby("movieId")["rating"].count().sort_values(ascending=Fal

top_movies.head(10)
```

Out[8]:
```
movieId
356     329
318     317
296     307
593     279
2571    278
260     251
480     238
110     237
589     224
527     220
Name: rating, dtype: int64
```

In [9]:
```
top_users = ratings.groupby("userId")["rating"].count().sort_values(ascending=False

top_users.head(10)
```

Out[9]:
```
userId
414    2698
599    2478
474    2108
448    1864
274    1346
610    1302
68     1260
380    1218
606    1115
288    1055
Name: rating, dtype: int64
```

In [10]:
```
data = pd.merge(ratings, movies, on="movieId")

data.head()
```

Out[10]:

	userId	movieId	rating	timestamp	title	genre
0	1	1	4.0	964982703	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantas
1	1	3	4.0	964981247	Grumpier Old Men (1995)	Comedy|Romanc
2	1	6	4.0	964982224	Heat (1995)	Action|Crime|Thrille
3	1	47	5.0	964983815	Seven (a.k.a. Se7en) (1995)	Mystery|Thrille
4	1	50	5.0	964982931	Usual Suspects, The (1995)	Crime|Mystery|Thrille

In [11]:
```
data = pd.merge(ratings, movies, on="movieId")

data.head()
```

Out[11]:

	userId	movieId	rating	timestamp	title	genre
0	1	1	4.0	964982703	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantas
1	1	3	4.0	964981247	Grumpier Old Men (1995)	Comedy|Romanc
2	1	6	4.0	964982224	Heat (1995)	Action|Crime|Thrille
3	1	47	5.0	964983815	Seven (a.k.a. Se7en) (1995)	Mystery|Thrille
4	1	50	5.0	964982931	Usual Suspects, The (1995)	Crime|Mystery|Thrille

In [12]:

```
top_rated = data.groupby("title")["rating"].mean().sort_values(ascending=False)

top_rated.head(10)
```

Out[12]:

```
title
Karlson Returns (1970)                                                    5.0
Zeitgeist: Moving Forward (2011)                                          5.0
Dream of Light (a.k.a. Quince Tree Sun, The) (Sol del membrillo, El) (1992)    5.0
Dragons: Gift of the Night Fury (2011)                                    5.0
12 Angry Men (1997)                                                       5.0
Justice League: Doom (2012)                                               5.0
Junior and Karlson (1968)                                                 5.0
Jump In! (2007)                                                           5.0
Human Condition III, The (Ningen no joken III) (1961)                     5.0
Louis Theroux: Law & Disorder (2008)                                      5.0
Name: rating, dtype: float64
```

In [13]:

```
#STEP 3: First Recommendation System (Basic Model)
```

In [14]:

```
user_movie_matrix = data.pivot_table(index='userId', columns='title', values='ratin

user_movie_matrix.head()
```

Out[14]:

title	'71 (2014)	'Hellboy': The Seeds of Creation (2004)	'Round Midnight (1986)	'Salem's Lot (2004)	'Til There Was You (1997)	'Tis the Season for Love (2015)	'burbs, The (1989)	'night Mother (1986)	(500) Days of Summer (2009)
userId									
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
5	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

5 rows × 9719 columns

In [15]:
```
movie_name = "Toy Story (1995)"

movie_ratings = user_movie_matrix[movie_name]
movie_ratings.head()
```

Out[15]:
```
userId
1    4.0
2    NaN
3    NaN
4    NaN
5    4.0
Name: Toy Story (1995), dtype: float64
```

In [16]:
```
similar_movies = user_movie_matrix.corrwith(movie_ratings)
```

```
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3015: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: invalid value encountered in multiply
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3023: RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3024: RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
```

In [17]:
```
corr_df = pd.DataFrame(similar_movies, columns=['Correlation'])
corr_df.dropna(inplace=True)

corr_df.head()
```

Out[17]:

title	Correlation
'burbs, The (1989)	0.240563
(500) Days of Summer (2009)	0.353833
***batteries not included (1987)**	-0.427425
10 Cent Pistol (2015)	1.000000
10 Cloverfield Lane (2016)	-0.285732

In [18]:
```
ratings_count = data.groupby('title')['rating'].count()

corr_df['num_ratings'] = ratings_count
```

In [19]:
```
recommendations = corr_df[corr_df['num_ratings'] > 100].sort_values('Correlation',

recommendations.head(10)
```

Out[19]:

title	Correlation	num_ratings
Toy Story (1995)	1.000000	215
Incredibles, The (2004)	0.643301	125
Finding Nemo (2003)	0.618701	141
Aladdin (1992)	0.611892	183
Monsters, Inc. (2001)	0.490231	132
Mrs. Doubtfire (1993)	0.446261	144
Amelie (Fabuleux destin d'Amélie Poulain, Le) (2001)	0.438237	120
American Pie (1999)	0.420117	103
Die Hard: With a Vengeance (1995)	0.410939	144
E.T. the Extra-Terrestrial (1982)	0.409216	122

In [20]:
```
def recommend_movies(movie_name, min_ratings=100):
    movie_ratings = user_movie_matrix[movie_name]

    similar_movies = user_movie_matrix.corrwith(movie_ratings)
    corr_df = pd.DataFrame(similar_movies, columns=['Correlation'])
    corr_df.dropna(inplace=True)
```

```
    ratings_count = data.groupby('title')['rating'].count()
    corr_df['num_ratings'] = ratings_count

    recommendations = corr_df[corr_df['num_ratings'] > min_ratings] \
                        .sort_values('Correlation', ascending=False)

    return recommendations.head(10)
```

In [21]:
```
recommend_movies("Toy Story (1995)")
```

```
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3015: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: invalid value encountered in multiply
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3023: RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3024: RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
```

Out[21]:

title	Correlation	num_ratings
Toy Story (1995)	1.000000	215
Incredibles, The (2004)	0.643301	125
Finding Nemo (2003)	0.618701	141
Aladdin (1992)	0.611892	183
Monsters, Inc. (2001)	0.490231	132
Mrs. Doubtfire (1993)	0.446261	144
Amelie (Fabuleux destin d'Amélie Poulain, Le) (2001)	0.438237	120
American Pie (1999)	0.420117	103
Die Hard: With a Vengeance (1995)	0.410939	144
E.T. the Extra-Terrestrial (1982)	0.409216	122

In [22]:
```
recommend_movies("Matrix, The (1999)")
```

```
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3015: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: invalid value encountered in multiply
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3023: RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3024: RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
```

Out[22]:

title	Correlation	num_ratings
Matrix, The (1999)	1.000000	278
Die Hard (1988)	0.544466	145
Inception (2010)	0.514767	143
Braveheart (1995)	0.496045	237
Aliens (1986)	0.470865	126
Lion King, The (1994)	0.444932	172
Monsters, Inc. (2001)	0.441205	132
Batman Begins (2005)	0.440338	116
Jurassic Park (1993)	0.427936	238
Fight Club (1999)	0.417196	218

In [23]:

```
#STEP 4: Matrix Factorization (SVD Model)
```

In [29]:

```
from sklearn.decomposition import TruncatedSVD

matrix = user_movie_matrix.fillna(0)

svd = TruncatedSVD(n_components=20)
latent_matrix = svd.fit_transform(matrix)

print(latent_matrix.shape)
```

```
(610, 20)
```

In [34]:

```
from sklearn.metrics.pairwise import cosine_similarity

similarity = cosine_similarity(latent_matrix)
similarity
```

Out[34]:
```
array([[ 1.        ,  0.11843342,  0.64136642, ...,  0.68660962,
         0.3295929 ,  0.22568157],
       [ 0.11843342,  1.        , -0.00179917, ...,  0.1585257 ,
         0.17905932,  0.27220499],
       [ 0.64136642, -0.00179917,  1.        , ...,  0.30217847,
         0.07208458,  0.28907127],
       ...,
       [ 0.68660962,  0.1585257 ,  0.30217847, ...,  1.        ,
         0.29611103,  0.3985442 ],
       [ 0.3295929 ,  0.17905932,  0.07208458, ...,  0.29611103,
         1.        ,  0.14006241],
       [ 0.22568157,  0.27220499,  0.28907127, ...,  0.3985442 ,
         0.14006241,  1.        ]], shape=(610, 610))
```

In [35]:
```
similarity_df = pd.DataFrame(similarity, index=user_movie_matrix.index)
```

In [40]:
```
def recommend_for_user(user_id, top_n=10, min_ratings=10):
    # 1) similar users
    similar_users = similarity_df[user_id].sort_values(ascending=False).drop(user_i
    top_users = similar_users.head(20).index  # use more neighbors

    # 2) candidate movies from similar users
    rec = data[data['userId'].isin(top_users)]

    # 3) remove already watched
    watched = data[data['userId'] == user_id]['title']
    rec = rec[~rec['title'].isin(watched)]

    # 4) aggregate
    agg = rec.groupby('title').agg(
        mean_rating=('rating', 'mean'),
        num_ratings=('rating', 'count')
    )

    # 5) filter low-support items
    agg = agg[agg['num_ratings'] >= min_ratings]

    # 6) weighted score (IMDb-style)
    C = data['rating'].mean()  # global mean
    m = min_ratings
    agg['score'] = (agg['num_ratings']/(agg['num_ratings']+m))*agg['mean_rating'] +
                   (m/(agg['num_ratings']+m))*C

    return agg.sort_values('score', ascending=False).head(top_n)
```

In [41]:
```
recommend_for_user(1)
```

Out[41]:

title	mean_rating	num_ratings	score
Inception (2010)	4.450000	20	4.133852
Shawshank Redemption, The (1994)	4.470588	17	4.111688
Departed, The (2006)	4.461538	13	4.044155
Dark Knight, The (2008)	4.333333	18	4.036270
Wolf of Wall Street, The (2013)	4.500000	11	4.024551
Lord of the Rings: The Return of the King, The (2003)	4.375000	12	3.977980
Shutter Island (2010)	4.318182	11	3.929313
Dark Knight Rises, The (2012)	4.250000	10	3.875778
Interstellar (2014)	4.153846	13	3.870242
Prestige, The (2006)	4.200000	10	3.850778

In [42]:
```
#STEP 5: Compare Models
```

In [43]:
```
corr_recs = recommend_movies("Toy Story (1995)").head(10)
svd_recs = recommend_for_user(1, min_ratings=10)

print("Correlation Model:\n", corr_recs.head(5))
print("\nSVD Model:\n", svd_recs.head(5))
```

```
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3015: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:2888: RuntimeWarning: invalid value encountered in multiply
  c *= np.true_divide(1, fact)
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3023: RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
C:\Users\girna\AppData\Local\Programs\Python\Python313\Lib\site-packages\numpy\lib\_
function_base_impl.py:3024: RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
```

```
Correlation Model:
                          Correlation  num_ratings
title
Toy Story (1995)            1.000000          215
Incredibles, The (2004)     0.643301          125
Finding Nemo (2003)         0.618701          141
Aladdin (1992)              0.611892          183
Monsters, Inc. (2001)       0.490231          132

SVD Model:
                                   mean_rating  num_ratings     score
title
Inception (2010)                     4.450000           20  4.133852
Shawshank Redemption, The (1994)     4.470588           17  4.111688
Departed, The (2006)                 4.461538           13  4.044155
Dark Knight, The (2008)              4.333333           18  4.036270
Wolf of Wall Street, The (2013)      4.500000           11  4.024551
```

In [44]:
```
#STEP 6: Evaluation
```

In [52]:
```
def precision_at_k(user_id, k=10):
    from sklearn.model_selection import train_test_split
    from sklearn.decomposition import TruncatedSVD
    from sklearn.metrics.pairwise import cosine_similarity

    # split FULL dataset
    train_data, test_data = train_test_split(data, test_size=0.2, random_state=42)

    # build matrix from train data
    train_matrix = train_data.pivot_table(index='userId', columns='title', values='

    # SVD
    svd = TruncatedSVD(n_components=20)
    latent = svd.fit_transform(train_matrix)

    # similarity
    similarity = cosine_similarity(latent)
    sim_df = pd.DataFrame(similarity, index=train_matrix.index, columns=train_matri

    # check user exists
    if user_id not in sim_df.index:
        return 0

    # find similar users
    similar_users = sim_df.loc[user_id].sort_values(ascending=False).drop(user_id)
    top_users = similar_users.head(10).index

    # recommendations from train data
    rec = train_data[train_data['userId'].isin(top_users)]

    # remove already watched
    watched = train_data[train_data['userId'] == user_id]['title']
    rec = rec[~rec['title'].isin(watched)]

    recs = rec.groupby('title')['rating'].mean().sort_values(ascending=False).head(
```

```
    # actual from test set
    actual = set(test_data[test_data['userId'] == user_id]['title'])
    recommended = set(recs)

    if len(actual) == 0:
        return 0

    return len(actual & recommended) / k
```

In [55]:
```
def avg_precision(k=10, n_users=20):
    users = data['userId'].unique()[:n_users]

    scores = []
    for u in users:
        score = precision_at_k(u, k)
        scores.append(score)

    return sum(scores) / len(scores)
```

In [56]:
```
avg_precision()
```

Out[56]: 0.015000000000000003

In [58]:
```
precision_at_k(1, k=20)
```

Out[58]: 0.05

In [59]:
```
#Build UI (Streamlit
```

In [62]:
```
!pip install streamlit
```

```
Requirement already satisfied: streamlit in .\AppData\Local\Programs\Python\Python31
3\Lib\site-packages (1.56.0)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=4.0 in .\AppData\Local\Pro
grams\Python\Python313\Lib\site-packages (from streamlit) (6.0.0)
Requirement already satisfied: blinker<2,>=1.5.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<8,>=5.5 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from streamlit) (7.0.5)
Requirement already satisfied: click<9,>=7.0 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from streamlit) (8.1.8)
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in .\AppData\Local\Progr
ams\Python\Python313\Lib\site-packages (from streamlit) (3.1.46)
Requirement already satisfied: numpy<3,>=1.23 in .\AppData\Local\Programs\Python\Pyt
hon313\Lib\site-packages (from streamlit) (2.4.2)
Requirement already satisfied: packaging>=20 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from streamlit) (24.2)
Requirement already satisfied: pandas<4,>=1.4.0 in .\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from streamlit) (3.0.1)
Requirement already satisfied: pillow<13,>=7.1.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from streamlit) (11.0.0)
Requirement already satisfied: pydeck<1,>=0.8.0b4 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from streamlit) (0.9.2)
Requirement already satisfied: protobuf<8,>=3.20 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from streamlit) (7.34.1)
Requirement already satisfied: pyarrow>=7.0 in .\AppData\Local\Programs\Python\Pytho
n313\Lib\site-packages (from streamlit) (23.0.1)
Requirement already satisfied: requests<3,>=2.27 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from streamlit) (2.32.3)
Requirement already satisfied: tenacity<10,>=8.1.0 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from streamlit) (9.1.4)
Requirement already satisfied: toml<2,>=0.10.1 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from streamlit) (0.10.2)
Requirement already satisfied: tornado!=6.5.0,<7,>=6.0.3 in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from streamlit) (6.4.1)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in .\AppData\Local\Progr
ams\Python\Python313\Lib\site-packages (from streamlit) (4.14.1)
Requirement already satisfied: watchdog<7,>=2.1.5 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from streamlit) (6.0.0)
Requirement already satisfied: jinja2 in .\AppData\Local\Programs\Python\Python313\L
ib\site-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (3.1.4)
Requirement already satisfied: jsonschema>=3.0 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (4.23.0)
Requirement already satisfied: narwhals>=1.27.1 in .\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (2.5.0)
Requirement already satisfied: colorama in .\AppData\Local\Programs\Python\Python313
\Lib\site-packages (from click<9,>=7.0->streamlit) (0.4.6)
Requirement already satisfied: gitdb<5,>=4.0.1 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from gitpython!=3.1.19,<4,>=3.0.7->streamlit) (4.0.12)
Requirement already satisfied: smmap<6,>=3.0.1 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from gitdb<5,>=4.0.1->gitpython!=3.1.19,<4,>=3.0.7->strea
mlit) (5.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in .\AppData\Local\Programs\Py
thon\Python313\Lib\site-packages (from pandas<4,>=1.4.0->streamlit) (2.9.0.post0)
Requirement already satisfied: tzdata in .\AppData\Local\Programs\Python\Python313\L
ib\site-packages (from pandas<4,>=1.4.0->streamlit) (2024.2)
Requirement already satisfied: charset-normalizer<4,>=2 in .\AppData\Local\Programs
```

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\Python\Python313\Lib\site-packages (from requests<3,>=2.27->streamlit) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in .\AppData\Local\Programs\Python\Pytho
n313\Lib\site-packages (from requests<3,>=2.27->streamlit) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from requests<3,>=2.27->streamlit) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from requests<3,>=2.27->streamlit) (2024.8.30)
Requirement already satisfied: MarkupSafe>=2.0 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from jinja2->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
(3.0.2)
Requirement already satisfied: attrs>=22.2.0 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->strea
mlit) (24.2.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in .\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from jsonschema>=3.0->altair!=5.4.
0,!=5.4.1,<7,>=4.0->streamlit) (2024.10.1)
Requirement already satisfied: referencing>=0.28.4 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0-
>streamlit) (0.35.1)
Requirement already satisfied: rpds-py>=0.7.1 in .\AppData\Local\Programs\Python\Pyt
hon313\Lib\site-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->stre
amlit) (0.21.0)
Requirement already satisfied: six>=1.5 in .\AppData\Local\Programs\Python\Python313
\Lib\site-packages (from python-dateutil>=2.8.2->pandas<4,>=1.4.0->streamlit) (1.16.
0)
```

In [63]:

```python
import streamlit as st

st.title("🎬 Movie Recommendation System")

user_id = st.number_input("Enter User ID", min_value=1, step=1)

if st.button("Recommend"):
    results = recommend_for_user(user_id, min_ratings=5)
    st.write(results)
```

```
2026-04-20 20:50:45.811 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.812 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.814 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.816 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.817 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.818 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.820 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.821 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.823 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.826 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.827 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.829 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.830 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.831 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.832 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2026-04-20 20:50:45.835 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
```

In [ ]:

In [ ]: